



Normal Microbiota of the GI Tract — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which region of the human body has the highest bacterial density?

EASY

- A. Stomach (10^2 – 10^3 /g)
- B. Duodenum (10^3 – 10^4 /mL)
- C. Ileum (10^6 – 10^7 /mL)
- D. Esophagus
- E. Cecum and transverse colon (10^{11} – 10^{12} /mL)

2. Which is the only organism that persists in the healthy human stomach?

EASY

- A. *Helicobacter pylori*
- B. *E. coli*
- C. *Lactobacillus*
- D. *B. fragilis*
- E. *C. difficile*

3. Which organisms dominate the intestinal flora of breast-fed children?

EASY

- A. Bifidobacteria
- B. *Lactobacilli*
- C. *Clostridia*
- D. *Enterobacterales*
- E. *Staphylococci*

4. The collective genomes of microbial symbionts are called the:

EASY

- A. Microbiome
- B. Microbiota
- C. Pathobiont
- D. Biofilm
- E. Virome

5. Why is stomach bacterial density only 10^2 – 10^3 per gram?

MEDIUM

- A. Rapid peristalsis
- B. Gastric enzymes and acidic pH
- C. High bile content
- D. High oxygen tension
- E. Lack of nutrients

6. A patient takes antacids and an H_2 -receptor antagonist for reflux. Which change is expected?

MEDIUM

- A. Fewer bacteria in the colon
- B. No change
- C. Sterile stomach
- D. Marked increase in gastric flora, including fecal organisms
- E. Selective growth of *H. pylori* only

7. A strict vegetarian and a meat-eater differ in fecal flora. Which is expected in the vegetarian?

MEDIUM

- A. ↑ Proteolytic *Clostridium*
- B. ↑ Bile-tolerant *Bilophila*
- C. Sterile stool
- D. ↑ Saccharolytic organisms such as *E. coli*
- E. ↑ *Alistipes*

8. A child has severe watery diarrhea. What happens to intestinal bacterial counts?

MEDIUM

- A. They rise sharply
- B. They fall as the tract is washed out
- C. They are unchanged
- D. They become sterile
- E. Only anaerobes rise

9. Which set lists three of the six predominant colonic phyla?

MEDIUM

- A. Bacteroidetes, Cyanobacteria, Firmicutes
- B. Bacteroidetes, Firmicutes, Actinobacteria
- C. Firmicutes, Ascomycota, Fusobacteria
- D. Proteobacteria, Cyanobacteria, Actinobacteria
- E. Bacteroidetes, Ascomycota, Cyanobacteria

10. Which archaeon is a methane producer in the colon?

MEDIUM

- A. *Bacteroides fragilis*
- B. *Candida albicans*
- C. *Bifidobacterium*
- D. *Enterococcus*
- E. *Methanobrevibacter smithii*

11. Which vitamins are synthesised by gut microbiota?

MEDIUM

- A. Vitamin K, biotin and folate
- B. Vitamin C, D and E
- C. Vitamin A and niacin only
- D. Vitamin B₁₂ only
- E. None

12. Which statement about transient microbiota is correct?

MEDIUM

- A. They always cause disease
- B. They are permanent
- C. They may colonise and cause disease if the resident flora is disturbed
- D. They are fixed at each site
- E. They are re-established after disturbance

13. How do commensals inhibit potential pathogens?

MEDIUM

- A. Producing gastric acid
- B. Destroying IgA
- C. Suppressing peristalsis
- D. Competing for nutrients and receptors and making bacteriocins and lactic acid
- E. Inducing bile release

14. Which pairing correctly matches the term to its meaning?

MEDIUM

- A. Symbiosis = imbalance; dysbiosis = stability
- B. Homeostasis = infection
- C. Symbiosis = interaction; homeostasis = stable result
- D. Dysbiosis = normal balance
- E. Symbiosis = stable state; homeostasis = interaction

15. A 70-year-old with diverticulosis has left-lower-quadrant pain and fever; an abscess is found next to the sigmoid. Which best explains the flora involved?

HARD

- A. A single *E. coli* strain
- B. *H. pylori* translocation
- C. Colonic anaerobes and other normal bowel flora after perforation
- D. Oral flora
- E. A transient environmental organism only

16. A hospitalised patient on broad-spectrum antibiotics develops severe recurrent colitis. Which treatment principle is described in the lecture?

HARD

- A. Prolonged more antibiotics only
- B. Fecal microbiota transplantation restoring Bacteroidetes and Firmicutes
- C. Antacids
- D. Bile acid depletion
- E. Probiotic yeast alone

17. Compared with a plant-based diet, an animal-based diet is associated with:

HARD

- A. ↑ Roseburia
- B. ↑ Ruminococcus bromii
- C. ↑ Firmicutes
- D. ↑ bile-tolerant Bacteroides and Alistipes; ↓ plant-polysaccharide-fermenting Firmicutes
- E. No change in flora

18. How can dysbiosis contribute to chronic systemic disease?

HARD

- A. Reduced IgA only
- B. Increased gastric acid
- C. Reduced fecal mass
- D. Increased vitamin K
- E. Increased permeability, endotoxin translocation and systemic inflammation

19. After a sigmoidoscopy a patient briefly has bacteria in the blood. What is the best explanation?

HARD

- A. Primary infection of the blood
- B. Failure of gastric acid
- C. Minor mucosal trauma allows transient bacteraemia from the dense colonic flora
- D. Overgrowth of *H. pylori*
- E. Permanent colonisation of blood

20. Which Bacteroides species is pathogenic through a toxin, and which suppresses C. difficile toxin?

HARD

- A. *B. thetaiotaomicron*; *B. fragilis*
- B. *B. ovatus*; *B. uniformis*
- C. *B. vulgatus*; *B. ovatus*
- D. *B. uniformis*; *B. fragilis*
- E. *B. fragilis* (BFT); *B. thetaiotaomicron*

✓ Answer key

Q	Answer	Difficulty	Why
1	E	EASY	The colon holds the highest count of any microbial habitat.
2	A	EASY	Of hundreds of phylotypes detected, only <i>H. pylori</i> persists.
3	A	EASY	Bottle-fed children have a more mixed flora.
4	A	EASY	Microbiota = organisms; microbiome = their genomes.
5	B	MEDIUM	Acid and enzymes limit bacterial numbers.
6	D	MEDIUM	Reduced acidity allows many organisms to survive.
7	D	MEDIUM	Vegetarian diets raise saccharolytic organisms; meat raises proteolytic ones.
8	B	MEDIUM	Counts fall in diarrhea and rise in stasis.
9	B	MEDIUM	The six are Bacteroidetes, Firmicutes, Actinobacteria, Verrucomicrobiota, Fusobacteria, Proteobacteria.
10	E	MEDIUM	It is low in abundance but may stabilise gut communities.
11	A	MEDIUM	The lecture lists vitamin K, biotin and folate.
12	C	MEDIUM	Transient organisms are usually insignificant while residents remain intact.
13	D	MEDIUM	These are the displacement mechanisms listed.
14	C	MEDIUM	Symbiosis describes the interaction; homeostasis the resulting stable condition.
15	C	HARD	Anaerobic flora (<i>B. fragilis</i> , clostridia, peptostreptococci) plays a main role in abscess formation after perforation.
16	B	HARD	FMT uses bacterial interference to displace <i>C. difficile</i> .
17	D	HARD	Animal-based diets shift toward bile-tolerant organisms.
18	E	HARD	The lecture figure links leaky gut and endotoxins to metabolic, cardiovascular, autoimmune and neurological disorders.
19	C	HARD	About 10% of such procedures cause transient bacteraemia.
20	E	HARD	<i>B. fragilis</i> toxin causes disease; <i>B. thetaiotaomicron</i> is anti-inflammatory and suppresses <i>C. diff</i> toxin production.